# JEE (ADVANCED) 2019 PAPER 1
## PART-II CHEMISTRY

**SECTION 1 (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct option is chosen; |
| *Zero Marks* | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| *Negative Marks* | : | –1 | In all other cases. |

Q.1 The green colour produced in the borax bead test of a chromium(III) salt is due to

(A) $Cr(BO_2)_3$ (B) $Cr_2(B_4O_7)_3$ (C) $Cr_2O_3$ (D) CrB

Q.2 Calamine, malachite, magnetite and cryolite, respectively, are

(A) $ZnSO_4$, $CuCO_3$, $Fe_2O_3$, $AlF_3$

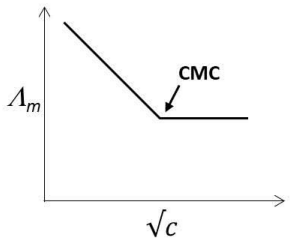
(B) $ZnSO_4$, $Cu(OH)_2$, $Fe_3O_4$, $Na_3AlF_6$

(C) $ZnCO_3$, $CuCO_3{\cdot}Cu(OH)_2$, $Fe_3O_4$, $Na_3AlF_6$

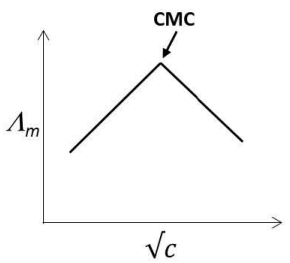
(D) $ZnCO_3$, $CuCO_3$, $Fe_2O_3$, $Na_3AlF_6$

Q.3 Molar conductivity ($\Lambda_m$) of aqueous solution of sodium stearate, which behaves as a strong electrolyte, is recorded at varying concentrations ($c$) of sodium stearate. Which one of the following plots provides the correct representation of micelle formation in the solution?
(critical micelle concentration (CMC) is marked with an arrow in the figures)

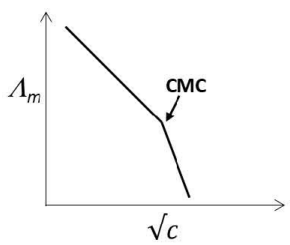
(A)



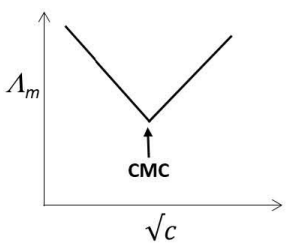

(B)



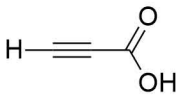

(C)



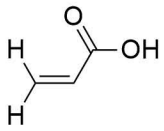

(D)



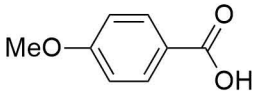
Q.4 The correct order of acid strength of the following carboxylic acids is



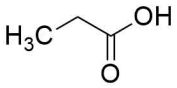

I




II




III




IV

(A) III > II > I > IV

(B) I > II > III > IV

(C) I > III > II > IV

(D) II > I > IV > III

**SECTION 2 (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : −1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get +4 marks;
  choosing ONLY (A) and (B) will get +2 marks;
  choosing ONLY (A) and (D) will get +2 marks;
  choosing ONLY (B) and (D) will get +2 marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (i.e. the question is unanswered) will get 0 marks; and
  choosing any other combination of options will get −1 mark.

Q.5 A tin chloride **Q** undergoes the following reactions (not balanced)

$\mathbf{Q} + Cl^- \rightarrow \mathbf{X}$
$\mathbf{Q} + Me_3N \rightarrow \mathbf{Y}$
$\mathbf{Q} + CuCl_2 \rightarrow \mathbf{Z} + CuCl$

X is a monoanion having pyramidal geometry. Both Y and Z are neutral compounds.
Choose the correct option(s)

(A) The central atom in X is $sp^3$ hybridized

(B) There is a coordinate bond in Y

(C) The oxidation state of the central atom in Z is +2

(D) The central atom in Z has one lone pair of electrons

Q.6 Fusion of $MnO_2$ with KOH in presence of $O_2$ produces a salt W. Alkaline solution of W upon electrolytic oxidation yields another salt X. The manganese containing ions present in W and X, respectively, are Y and Z. Correct statement(s) is(are)

(A) In aqueous acidic solution, Y undergoes disproportionation reaction to give Z and $MnO_2$

(B) Both Y and Z are coloured and have tetrahedral shape

(C) Y is diamagnetic in nature while Z is paramagnetic

(D) In both Y and Z, $\pi$-bonding occurs between $p$-orbitals of oxygen and $d$-orbitals of manganese

Q.7 Choose the reaction(s) from the following options, for which the standard enthalpy of reaction is equal to the standard enthalpy of formation.

(A) $2H_2(g) + O_2(g) \rightarrow 2H_2O\ (l)$

(B) $2C(g) + 3H_2(g) \rightarrow C_2H_6(g)$

(C) $\frac{3}{2}O_2(g) \rightarrow O_3(g)$

(D) $\frac{1}{8}S_8(s) + O_2(g) \rightarrow SO_2(g)$

Q.8 Which of the following statement(s) is(are) correct regarding the root mean square speed ($u_{rms}$) and average translational kinetic energy ($\varepsilon_{av}$) of a molecule in a gas at equilibrium ?

(A) $u_{rms}$ is doubled when its temperature is increased four times

(B) $\varepsilon_{av}$ is doubled when its temperature is increased four times

(C) $\varepsilon_{av}$ at a given temperature **does not** depend on its molecular mass

(D) $u_{rms}$ is inversely proportional to the square root of its molecular mass

Q.9 Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature.

(A) $BeCl_2$, $CO_2$, $BCl_3$, $CHCl_3$

(B) $NO_2$, $NH_3$, $POCl_3$, $CH_3Cl$

(C) $BF_3$, $O_3$, $SF_6$, $XeF_6$

(D) $SO_2$, $C_6H_5Cl$, $H_2Se$, $BrF_5$

Q.10 In the decay sequence,

$$^{238}_{92}U \xrightarrow{-x_1} {}^{234}_{90}Th \xrightarrow{-x_2} {}^{234}_{91}Pa \xrightarrow{-x_3} {}^{234}\mathbf{Z} \xrightarrow{-x_4} {}^{230}_{90}Th$$

$x_1, x_2, x_3$ and $x_4$ are particles/radiation emitted by the respective isotopes. The correct option(s) is(are)

(A) $x_1$ will deflect towards negatively charged plate

(B) $x_2$ is $\beta^-$

(C) $x_3$ is $\gamma$-ray

(D) **Z** is an isotope of uranium

Q.11 Which of the following statement(s) is(are) true?

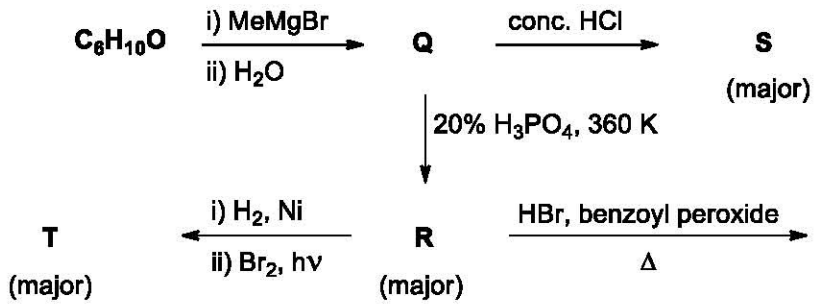
(A) Monosaccharides **cannot** be hydrolysed to give polyhydroxy aldehydes and ketones

(B) Oxidation of glucose with bromine water gives glutamic acid

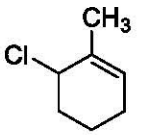
(C) Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose

(D) The two six-membered cyclic hemiacetal forms of D-(+)-glucose are called anomers

Q.12 Choose the correct option(s) for the following set of reactions




(A)




S

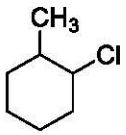



T

(B)

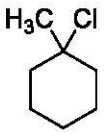



U




S

(C)




S




U

(D)




U




T

**SECTION 3 (Maximum Marks: 18)**

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  *Full Marks* : +3 If ONLY the correct numerical value is entered;
  *Zero Marks* : 0 In all other cases.

Q.13 Among $B_2H_6$, $B_3N_3H_6$, $N_2O$, $N_2O_4$, $H_2S_2O_3$ and $H_2S_2O_8$, the total number of molecules containing covalent bond between two atoms of the same kind is_____

Q.14 At 143 K, the reaction of $XeF_4$ with $O_2F_2$ produces a xenon compound Y. The total number of lone pair(s) of electrons present on the whole molecule of Y is_____

Q.15 For the following reaction, the equilibrium constant $K_c$ at 298 K is $1.6 \times 10^{17}$.

$$Fe^{2+}(aq) + S^{2-}(aq) \rightleftharpoons FeS(s)$$

When equal volumes of 0.06 M $Fe^{2+}(aq)$ and 0.2 M $S^{2-}(aq)$ solutions are mixed, the equilibrium concentration of $Fe^{2+}(aq)$ is found to be $Y \times 10^{-17}$ M. The value of Y is _____

Q.16 On dissolving 0.5 g of a non-volatile non-ionic solute to 39 g of benzene, its vapor pressure decreases from 650 mm Hg to 640 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is _____
(Given data: Molar mass and the molal freezing point depression constant of benzene are 78 g $mol^{-1}$ and 5.12 K kg $mol^{-1}$, respectively)

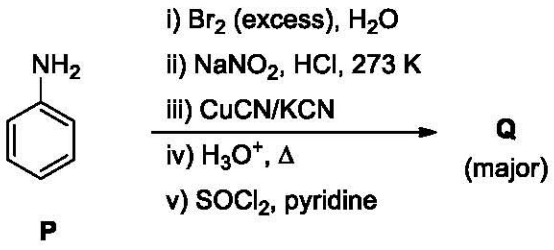
Q.17 Consider the kinetic data given in the following table for the reaction A + B + C → Product.

| Experiment No. | [A] (mol $dm^{-3}$) | [B] (mol $dm^{-3}$) | [C] (mol $dm^{-3}$) | Rate of reaction (mol $dm^{-3}$ $s^{-1}$) |
|---|---|---|---|---|
| 1 | 0.2 | 0.1 | 0.1 | $6.0 \times 10^{-5}$ |
| 2 | 0.2 | 0.2 | 0.1 | $6.0 \times 10^{-5}$ |
| 3 | 0.2 | 0.1 | 0.2 | $1.2 \times 10^{-4}$ |
| 4 | 0.3 | 0.1 | 0.1 | $9.0 \times 10^{-5}$ |

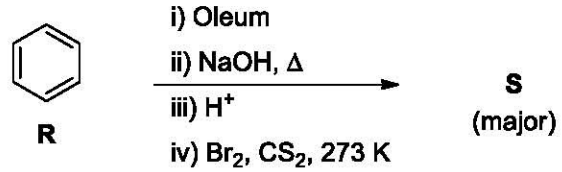
The rate of the reaction for [A] = 0.15 mol $dm^{-3}$, [B] = 0.25 mol $dm^{-3}$ and [C] = 0.15 mol $dm^{-3}$ is found to be $Y \times 10^{-5}$ mol $dm^{-3}$ $s^{-1}$. The value of Y is_____

Q.18 Schemes 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is _____

**Scheme 1:**




**Scheme 2:**




**Scheme 3:**

